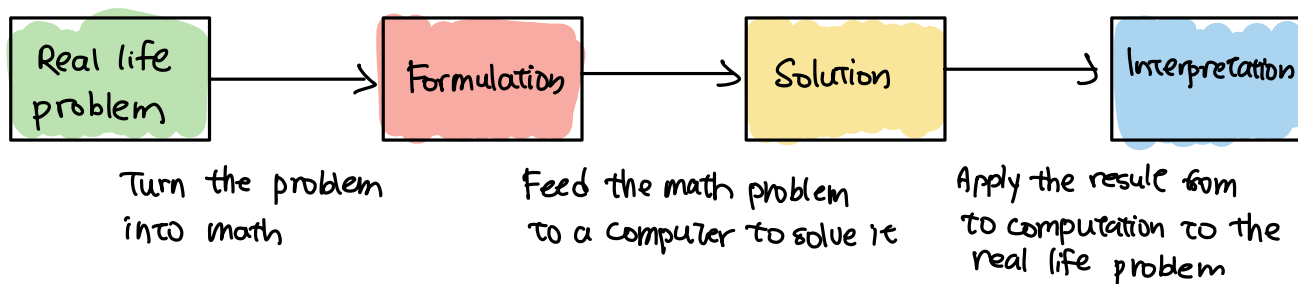


CO25D Review 1: LP formulations

Optimization: Best way to do something, e.g. how to make the most money, how to operate a company efficiently, etc.

Optimization work flow



Focus of the first module: formulation.

Formulation

3 main steps in formulating a problem:

- ① Define variables. "what are we trying to decide?"
- ② Objective function. $\max f(x)$ or $\min f(x)$. "what are we trying to optimize?"
- ③ Constraints. $g(x) \leq b$, $g(x) = b$, $g(x) \geq b$. "what limits the values of the variables?"

First week: formulate as linear programs, i.e. $f(x), g(x) \dots$ are linear/affine, e.g.

$$3x_1 + 4x_2 - 5x_3 + 2.$$

Example:

Suppose we are selling apples and bananas at a stand. Apples sell for \$2 per kilogram, and bananas sell for \$1.5 per kilogram. Our stand holds up to 75 kilogram of fruits. Also, there are only 4 square metres of shelf space. Each kilogram of apples / bananas takes up roughly 0.08 / 0.05 square metres of shelf space, respectively. How much of each fruit should we stock to maximize the total sales?

Variables: Let x_1 be the amount of apples (in kg) that we sell.

Let x_2 " " bananas " " .

Objective function: $\max 2x_1 + 1.5x_2$

Constraints: Weight constraint: $x_1 + x_2 \leq 75$

Space constraint: $0.08x_1 + 0.05x_2 \leq 4$

Non-negativity: $x_1, x_2 \geq 0$

Optimal solution: $x_1 = 8\frac{1}{3}$, $x_2 = 66\frac{2}{3}$, total value $\$116\frac{2}{3}$.

Multiperiod model

Things happen over some periods of time. For each type of variable, we have multiple copies of it depending on the time period.

Balance constraints: for each time period, keep track of where things are.

"things coming in = things going out"